

## Parent Functions II: $x^3$ and $\sqrt[3]{x}$

**TEACHER KEY** | Algebra II | Unit 1 | Day 5 | TEK A2.2.A

**Objective:** I can graph the cubic and cube root parent functions from a table and analyze their key attributes, including their rotational symmetry about the origin.

### Vocabulary

**Rotational symmetry** — turning the graph  **$180^\circ$**  about the origin lands it back on itself.

**Reflectional symmetry** — the graph can be **folded** across a line onto itself.

**Cube root** —  $\sqrt[3]{a}$  asks: what number **cubed** gives  $a$ ? Negatives **are** allowed, unlike square roots.

### Part 1 — The Cubic Function, $f(x) = x^3$

Fill in the table. Notice what a negative input produces.

|                                |    |    |   |   |   |
|--------------------------------|----|----|---|---|---|
| <b>x</b>                       | -2 | -1 | 0 | 1 | 2 |
| <b><math>f(x) = x^3</math></b> | -8 | -1 | 0 | 1 | 8 |

A negative input gives a **negative** output, because a negative cubed stays negative.

### Part 2 — The Cube Root Function, $f(x) = \sqrt[3]{x}$

This undoes the cubic. Compare it to your Part 1 table.

|                                        |    |    |   |   |   |
|----------------------------------------|----|----|---|---|---|
| <b>x</b>                               | -8 | -1 | 0 | 1 | 8 |
| <b><math>f(x) = \sqrt[3]{x}</math></b> | -2 | -1 | 0 | 1 | 2 |

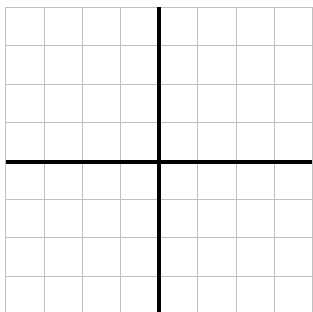
What do you notice next to Part 1? **the x and y values trade places**

We will come back to that on Day 10 - these two are **inverses**.

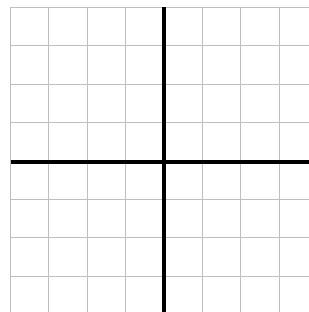
### Part 3 — Graph Them Both

Each square is 1 unit. Both grids run -8 to 8 on each axis, so 1 square = 2 units.

**A.  $f(x) = x^3$**  steep S-curve



**B.  $f(x) = \sqrt[3]{x}$**  flatter S-curve



Passes through the **origin**

Mirror line between them:  **$y = x$**

### Part 4 — Compare the Key Attributes

| Attribute         | <b><math>f(x) = x^3</math></b>        | <b><math>f(x) = \sqrt[3]{x}</math></b> |
|-------------------|---------------------------------------|----------------------------------------|
| Domain (interval) | <b><math>(-\infty, \infty)</math></b> | <b><math>(-\infty, \infty)</math></b>  |

|                         |                         |                         |
|-------------------------|-------------------------|-------------------------|
| Range (interval)        | $(-\infty, \infty)$     | $(-\infty, \infty)$     |
| x- and y-intercept      | $(0, 0)$                | $(0, 0)$                |
| Symmetry                | rotational, $180^\circ$ | rotational, $180^\circ$ |
| Maximum or minimum?     | neither                 | neither                 |
| Increasing / decreasing | always increasing       | always increasing       |

These two share every attribute above. The difference is the **steepness** of the curve.

## Part 5 — Compare to Day 4

1. Why does  $\sqrt[3]{x}$  accept negatives when  $\sqrt{x}$  does not? **a negative cubed is negative, but no real number squared is negative**
2. Which of the four functions so far have a minimum?  **$\sqrt{x}$  and  $|x|$**

## Practice

*Evaluate and explain. Show your work where there is room.*

1. Evaluate:  $(-3)^3 = -27$     $\sqrt[3]{-27} = -3$     $\sqrt[3]{64} = 4$     $\sqrt[3]{0} = 0$
2. Solve  $x^3 = 125$ .  $x = 5$    How many real answers? **1**
3. Solve  $x^2 = 25$ .  $x = 5$  and  $-5$    How many real answers? **2**
4. Items 2 and 3 give different counts. Explain why cubing behaves differently from squaring.  
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5.  $f(x) = x^3$  on the interval  $[-1, 3]$ : minimum **-1** maximum **27**
6. True or false: every real number has exactly one real cube root. **true**
7. Which of the four parent functions so far accept negative inputs?  
 **$|x|$ ,  $x^3$ , and  $\sqrt[3]{x}$  (not  $\sqrt{x}$ )**

## Exit Ticket

A function has domain and range both  $(-\infty, \infty)$ , passes through the origin, has  $180^\circ$  rotational symmetry, and no maximum or minimum.

Name two parent functions this could be:  **$x^3$  and  $\sqrt[3]{x}$**

What extra information would tell them apart? **a point such as  $(2, 8)$  versus  $(8, 2)$**